

2.5 21st Century Challenges

Geography — Eduqas A-Level (A110QS) — Component 2: Synoptic Theme

1. Synopticity — What Topic 2.5 Is and How It Is Examined

Topic 2.5 is the **synoptic theme** of the Eduqas A-level (A110QS) specification. It is not a new body of content but rather a lens through which candidates are expected to *integrate* material from across the specification — and especially from at least two of the four themes in Components 1 and 2 (changing landscapes, changing places, water and carbon cycles, and global governance — migration and the oceans), with credit for relevant links to Component 3 — to evaluate how the major geographical challenges of the twenty-first century are produced, distributed and contested. Synopticity rewards candidates who connect physical and human geography, who move fluently between local, regional, national and global scales, and who use named examples across multiple regions of the world rather than relying on a single favoured case study.

The Eduqas examination style for 2.5 emphasises **extended evaluative writing**. Typical questions take the form: “To what extent is climate change the most serious 21st-century challenge?” or “Assess the effectiveness of global governance in managing contemporary challenges.” The Section C question (Component 2) is a single 30-mark extended response, chosen from two questions and based on resource material; it is expected to produce a clear *thesis*, weigh multiple positions with located evidence, introduce *counter-arguments* and *alternative interpretations*, and reach a defended *conclusion*. UK-specific angles on climate leadership, post-Brexit trade or sustainable development (the Well-being of Future Generations (Wales) Act 2015 is a rare example of SDG-style goals being written into law) can add useful located detail where relevant.

Synopticity: The explicit drawing together of material from different parts of a specification so that candidates can demonstrate they understand geography as an *integrated discipline* — linking physical processes, human systems, scales and case studies — rather than as a set of isolated topics.

Exam tip — synopticity in practice: In any 30-mark Section C answer, signpost the synoptic links explicitly. Use phrases such as “linking to Topic 2.1”, “as seen in the 2.4 ocean governance material” or “the 2.3 migration case of Syria illustrates this”. Examiners cannot award synoptic credit for connections they cannot see you making.

2. Globalisation and Interdependence

The defining context of 21st-century geography is **globalisation**: the deepening of economic, political, social, environmental and technological connections between places. Globalisation is not new — the Silk Roads, the Atlantic slave trade and the British Empire were all globalising systems — but its *velocity, density and depth* since the 1990s are qualitatively new. A 2008 crisis in US sub-prime mortgages collapsed Icelandic banks; a 2020 virus in Wuhan halted factories in northern Italy; a 2022 war in Ukraine drove wheat prices in Egypt and petrol prices in Pembrokeshire. Interdependence is no longer a theory; it is daily experience.

Globalisation: The process by which national economies, cultures, institutions and populations have become increasingly interconnected through cross-border flows of goods, services, capital, labour, technology, ideas and environmental impacts, producing a degree of *interdependence* such that events in one place materially shape conditions in others.

Waves and phases of globalisation

Economic historians commonly identify three waves:

- **First wave (c. 1870–1914):** driven by steam shipping, the telegraph, the gold standard and liberal trade; ended by WWI and the inter-war collapse.
- **Second wave (c. 1945–1990):** Bretton Woods, GATT tariff cuts, containerisation (1956 onwards — reducing port handling costs ~20×), jet aviation. Still largely an OECD-to-OECD phenomenon.
- **Third wave (c. 1990–2008):** hyperglobalisation; collapse of the USSR; China’s WTO accession (2001); ICT revolution. Global trade grew ~2× faster than GDP.
- **Post-2008 “slowbalisation” or fourth phase:** global trade-to-GDP has plateaued; geopolitical rivalry, “friendshoring” and industrial policy are reshaping supply chains. The IMF documented rising trade restrictions, with ~3,200 new measures in 2023 alone.

Dimensions of globalisation

- **Economic:** global supply chains (an iPhone is designed in Cupertino, components drawn from 43 countries, assembled in Zhengzhou, sold globally), foreign direct investment (FDI), transnational corporations (TNCs). *Walmart* (revenue ~\$650 bn), *Amazon*, *Apple* (market capitalisation peaking >\$3 trillion in 2024), *Toyota* and *Saudi Aramco* are examples whose revenues exceed the GDP of most states.
- **Financial:** global capital flows (~\$7 trillion daily on foreign-exchange markets), currency interdependence, the 2008 Global Financial Crisis which began in US mortgage-backed securities and caused recessions across the OECD.
- **Political:** multilateral institutions — United Nations, World Trade Organization (WTO), International Monetary Fund (IMF), World Bank — and regional blocs (EU, ASEAN, African Union, Mercosur).
- **Social / cultural:** migration (Topic 2.3), global media, diaspora communities, the spread of English, hybridisation of food, music, dress.
- **Environmental:** the shared atmosphere, oceans (Topic 2.4) and biodiversity produce global externalities — climate change is the defining case.
- **Technological:** the internet (~5.4 bn users by 2024), undersea fibre-optic cables carrying ~99% of intercontinental data, satellites, AI foundation models released as global products.

The Brandt Line and its critique

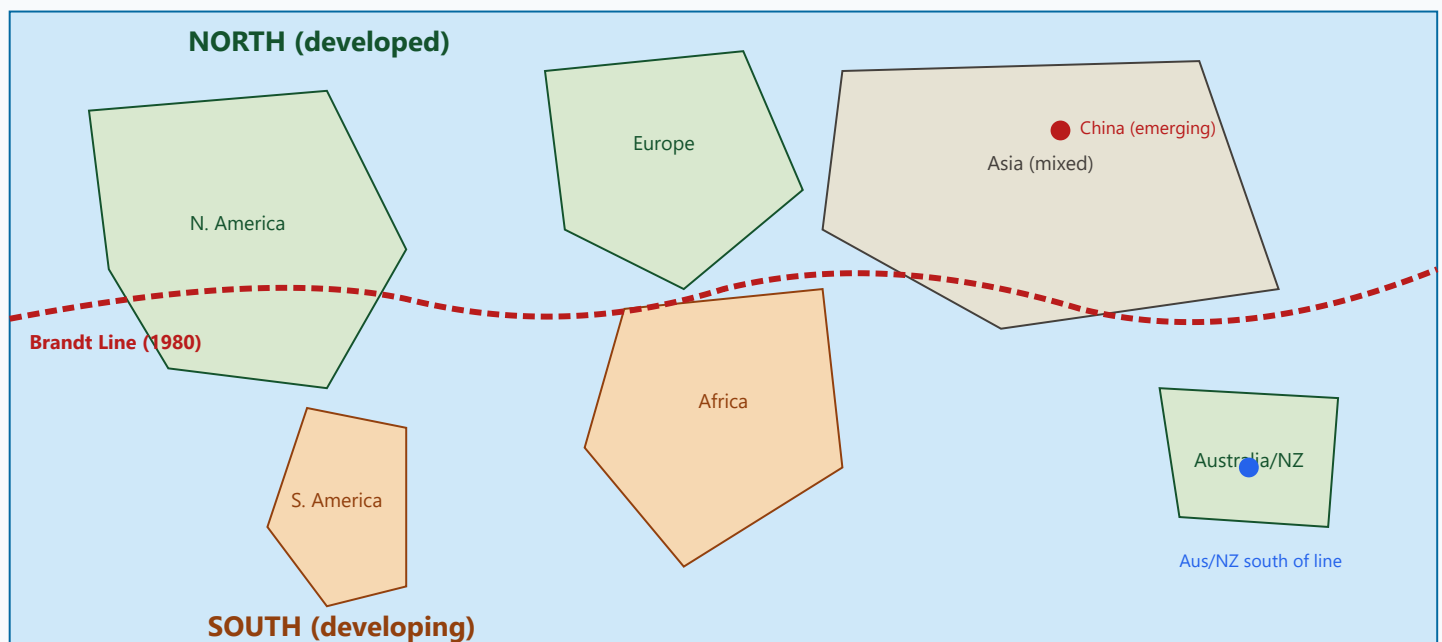
Willy Brandt's 1980 Commission divided the world schematically into a wealthy global **North** and a poor global **South**, with the line running south of the USA and Europe, north of most of Latin America, Africa and South Asia, but dipping below Australia and New Zealand. The line still has pedagogical value but has been repeatedly critiqued. China's GDP per capita rose from ~\$195 in 1980 to ~\$12,600 in 2023; South Korea, Singapore and the UAE are indisputably "developed"; meanwhile Detroit, eastern Kentucky and south Wales contain pockets of deprivation comparable to parts of the global South. The 21st-century map is *patchwork*, not banded.

Alternative framings include Hans Rosling's "four levels" (dollar-a-day, \$4/day, \$16/day, \$64/day) which emphasise that *most* of humanity now lives in the middle two levels rather than in either extreme; the World Bank's classification into Low-, Lower-Middle-, Upper-Middle- and High-Income economies; and Wallerstein's world-systems model of core, semi-periphery and periphery. Each has analytical strengths: Rosling's emphasises the *shape* of the global income distribution rather than a binary, the World Bank's is operationally tied to lending terms, Wallerstein's foregrounds historical and structural relations rather than

income snapshots. For Eduqas 2.5 essays, the key move is to *choose* a framing self-consciously rather than inheriting one.

Brandt Line (1980): A schematic North–South division of the world proposed by the Independent Commission on International Development Issues (chaired by former West German Chancellor Willy Brandt). Useful as a teaching device but increasingly criticised for obscuring the rise of emerging economies and intra-national inequality.

The Brandt Line (1980) — simplified world



Exam tip — always name examples: Questions on globalisation will be marked down for vague “developing countries” or “big companies”. Name Vietnam’s garment sector, name TSMC’s Taiwan fabs, name the Port of Rotterdam. A named example is the minimum bar; located, dated and quantified detail is what lifts an answer into the higher bands.

3. Superpowers and Shifting Geopolitics

Twentieth-century geopolitics was structured by a bipolar Cold War (USA vs USSR, 1947–1991) followed by a brief unipolar moment of American primacy. The early 21st century is increasingly **multipolar**: a world of several centres of economic and political power, with shifting alignments rather than fixed blocs.

Superpower: A state possessing overwhelming capability across multiple dimensions — military, economic, political, cultural, technological — allowing it to project influence globally. Joseph Nye distinguishes *hard power* (coercion, military, economic sanctions) from *soft power* (attraction, culture, values) and *smart power* (their combination).

Multipolar world: A geopolitical configuration in which three or more states or state groupings possess comparable power and no single hegemon dominates. Contrasted with *unipolar* (single hegemon) and *bipolar* (two rival superpowers) systems.

Geopolitics: The study of how geography (location, resources, territory) shapes the political behaviour of states, and how power politics shape geographical outcomes. Classical geopolitics (Mackinder, Spykman) emphasised land versus sea power; contemporary geopolitics adds cyberspace, supply-chain chokepoints, and resource geopolitics (rare earths, lithium, semiconductors).

Power is measured in several dimensions simultaneously. *Hard power* captures military and economic coercion. *Soft power* (Nye, 1990) captures attraction and agenda-setting through culture, values and institutions — Hollywood, K-pop, the English Premier League, Oxbridge, the Confucius Institutes. *Sharp power* (Walker and Ludwig, 2017) captures coercive influence operations short of war — disinformation, bribery, cyberattack. *Structural power* (Strange, 1988) captures the ability to shape the systems within which others operate — the dollar-denominated global financial system, the SWIFT messaging network, the US-dominated internet stack, the standards baked into the International Maritime Organization. Twenty-first-century geopolitics is increasingly contested on structural-power terrain: the question is not just who has the most tanks but who sets the operating system of the world.

The main poles in 2026

- **United States:** still the largest economy in nominal GDP (~\$28 tn, 2024), the world reserve currency (US dollar used in ~88% of FX transactions), the largest military spender (~\$997 bn in 2024 — ~37% of the global total, SIPRI), and the dominant power in AI, finance (Wall Street), and culture (Hollywood, Silicon Valley platforms). Leadership in “Big Tech” (Alphabet, Apple, Meta, Microsoft, Amazon, Nvidia).
- **China:** second-largest economy nominally (~\$18 tn) and largest on purchasing-power parity (PPP) terms; the “workshop of the world” responsible for ~28% of global manufacturing output. Belt and

Road Initiative since 2013 (see case study). Technology leadership in 5G (Huawei), EVs (BYD outsold Tesla globally in Q4 2023), solar PV and batteries. People's Liberation Army modernising rapidly.

- **European Union:** 27 member states, single market of ~450 million people, collectively the second- or third-largest economy depending on measure. Leader in regulatory power ("Brussels effect": GDPR, DSA, AI Act).
- **India:** most populous country since 2023 (~1.43 bn), fastest-growing large economy, demographic dividend (median age ~28), strong service sector (IT offshoring to Bangalore, Hyderabad). Fourth-largest economy nominally on IMF estimates for 2025 (overtaking Japan), projected third by ~2030.
- **Russia:** fossil-fuel power, the world's second military by strength and first by nuclear stockpile, but economy smaller than Italy's. Declining demographically. Post-2022 pariah in the West but enduring partnerships with China, India, Iran.
- **Others:** Japan, the UK, Brazil, South Africa, Saudi Arabia, Turkey, Indonesia form a second tier; BRICS bloc (Brazil, Russia, India, China, South Africa, joined in 2024 by the UAE, Iran, Egypt and Ethiopia and in 2025 by Indonesia) explicitly positions itself as a non-Western counterweight.

BRICS: Originally an acronym coined by Jim O'Neill (Goldman Sachs, 2001) for Brazil, Russia, India, China; later institutionalised with South Africa (2010) and expanded in 2024 to include the UAE, Iran, Egypt and Ethiopia, and in January 2025 Indonesia ("BRICS+"). Represents ~45% of the world's population and ~35% of global GDP (PPP). Positioned as an alternative to G7-dominated global governance.

Comparison of leading powers

Indicator	USA	China	EU-27	India	Russia
Population (m)	~335	~1,410	~450	~1,430	~144
GDP nominal (\$tn, 2024)	~28	~18	~19	~4.1	~2.0
GDP (PPP, \$tn)	~28	~35	~25	~14	~5.5
Military spend (\$bn, 2023)	916	296	~340 (combined)	84	109
Nuclear warheads (est.)	~5,044	~500	UK 225 + FR 290	~172	~5,580

HDI (2022, UNDP)	0.927	0.788	~0.94 avg	0.644	0.821
Emissions (GtCO ₂ , 2023)	~4.9	~11.9	~2.6	~2.9	~1.7

Exam tip — nominal vs PPP GDP: Specify which measure you are using. On *nominal* GDP the US is ahead of China; on *purchasing-power parity* China overtook the US around 2014 (IMF). The choice matters for arguments about economic “leadership”. Strong answers show this awareness.

4. Development and Inequality

Development is itself a contested concept. Modernisation theorists (Rostow, 1960) saw it as a linear path from traditional to high-mass-consumption society; dependency theorists (Frank, Wallerstein) argued that “underdevelopment” is produced by the *relations* between core and periphery rather than being a stage that simply “comes later”. Amartya Sen reframed development as the expansion of *capabilities and freedoms*, a move institutionalised in the UNDP’s Human Development Index.

Measures of development

Human Development Index (HDI): A composite index published by UNDP since 1990 combining life expectancy at birth, mean and expected years of schooling, and logged GNI per capita (PPP). Ranges 0–1; classified Very High (>0.8), High, Medium, Low. Norway, Switzerland and Iceland consistently rank at the top; South Sudan, Somalia, Chad at the bottom.

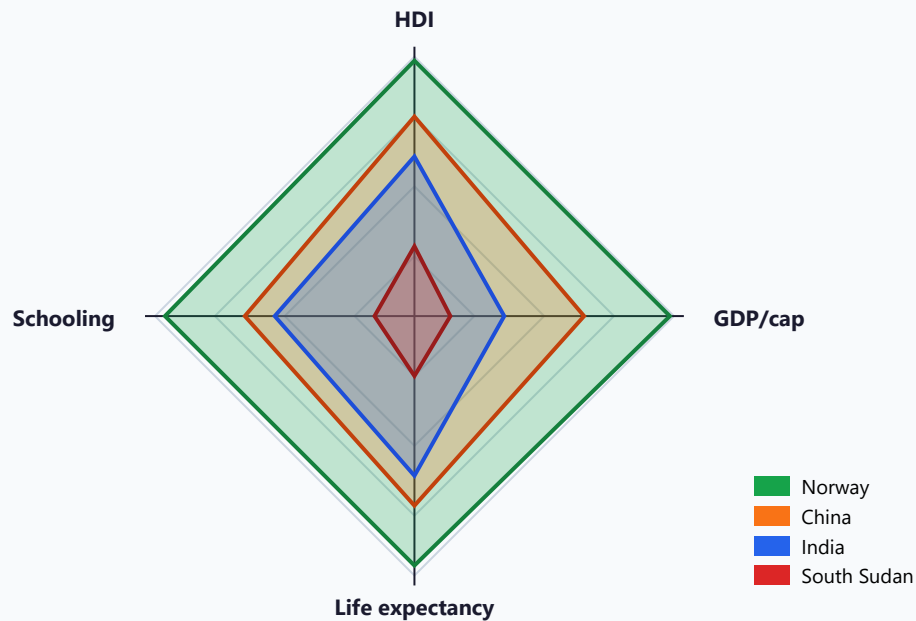
Gini coefficient: A statistical measure of income (or wealth) inequality ranging from 0 (perfect equality) to 1 (one household owns everything). Typical values: Scandinavia ~0.25–0.30, UK ~0.35, USA ~0.40, South Africa ~0.63 (highest among large economies). Geographers also use a *Lorenz curve* to visualise the distribution.

Measure	Body / year	What it captures	Limitations
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GDP per capita	IMF / World Bank	Average economic output	Ignores distribution, informal economy, non-market work
GNI per capita (Atlas/PPP)	World Bank	Income accruing to residents	Same as GDP plus expatriate flows
HDI	UNDP (since 1990)	Health + education + income	National average; no environment/governance dimension
Gini coefficient	World Bank	Inequality of distribution	Single number; can hide regional variation
Multidimensional Poverty Index (MPI)	OPHI / UNDP (2010)	Deprivations across 10 indicators in health, education, living standards	Data availability patchy
Gender Inequality Index (GII)	UNDP	Reproductive health, empowerment, labour market	Data gaps in low-income states
Happy Planet Index	NEF (2006)	Well-being + life expectancy / ecological footprint	Treated sceptically by economists

Development diamond — four countries compared

axes: HDI, GDP/cap (log), life expectancy, mean years schooling (each normalised 0–1)



Global inequality

Between-country inequality has narrowed slightly (thanks largely to China's catch-up) but *within-country* inequality has widened almost everywhere. UBS's *Global Wealth Report 2025* — the successor to the Credit Suisse series, which UBS absorbed in 2023 — estimated that the richest 1.5% of adults, those worth more than US\$1 million, owned **48.4% of all global household wealth**; the World Inequality Report 2026 finds the top 0.001% alone now hold around three times the wealth of the poorest half of humanity combined. Oxfam reported in January 2024 that the five richest men had more than doubled their fortunes since 2020, while the bottom five billion had grown poorer. Inequality matters for governance because it shapes political legitimacy, populism and migration pressures.

The *elephant graph* (Lakner and Milanovic, 2013) captures this striking pattern: between 1988 and 2008, the biggest gains in real income went to the global middle class (largely the emerging middle class in China) and to the very top of the distribution (the global 1%), while the upper-middle deciles of rich countries — the industrial working classes of the US and Europe — experienced near-stagnation. This "twin peaks" pattern has been interpreted as a structural cause of political backlash in advanced economies (Brexit, Trump, populism in France, Germany, Italy). The geography of inequality also has a spatial dimension: *cities* and *regions* diverge within countries. London's GDP per capita is ~2.7× that of north-east England; San Francisco's is ~2× that of Mississippi. The "levelling up" agenda in the UK and the Biden administration's "place-based" industrial policy (CHIPS Act, Inflation Reduction Act) are responses to this geography.

Sustainable Development Goals (SDGs): Seventeen goals adopted by all UN member states in September 2015 (Agenda 2030), succeeding the Millennium Development Goals. Cover poverty (SDG1), hunger (SDG2), health (SDG3), education (SDG4), gender (SDG5), water (SDG6), energy (SDG7), climate (SDG13), oceans (SDG14), life on land (SDG15), peace (SDG16) and partnerships (SDG17). Monitored through 231 indicators; progress seriously off-track in 2024 (UN estimate: only ~15% of targets on track).

Exam tip — SDGs as synoptic hinge: The SDGs are a gift for synoptic answers because each goal maps to a topic: SDG6 water (Topic 2.1), SDG7 energy (2.2), SDG10 reduced inequalities (migration 2.3), SDG14 oceans (2.4). Use them as a structural device for essays that ask about “21st-century challenges” generally.

5. Migration in the 21st Century (link to Topic 2.3)

As Topic 2.3 makes clear, human mobility has reached unprecedented levels — UN DESA estimated ~281 million international migrants in 2020 (~3.6% of global population), plus ~108 million forcibly displaced by mid-2023 (UNHCR). Migration is simultaneously a symptom (of war, climate, inequality) and a driver (of labour markets, demographic balance, cultural change) of 21st-century challenges.

- **Labour mobility and remittances:** remittances to low- and middle-income countries reached ~\$656 bn in 2023 (World Bank) — more than FDI plus aid combined. For Tonga, remittances are ~41% of GDP; Nepal ~23%; Tajikistan ~48%.
- **Brain drain / brain gain:** Zambia loses ~50% of its trained doctors to OECD economies; conversely the Indian diaspora in the USA (~5 million) reinvests in Bangalore’s IT cluster.
- **Refugees and asylum seekers** are indicators of political instability. Syria, Venezuela, Ukraine, Afghanistan and Sudan together produced >50% of the world’s refugees by 2024.
- **Ageing in the global North** — Japan’s median age is 49.5, Germany’s 46, Italy’s 48. These societies face shrinking workforces and rising dependency ratios, which create structural demand for immigration even where politics resists it.
- **Population growth in Africa** — Nigeria is projected to reach ~360 million by 2050; the Democratic Republic of Congo and Tanzania also triple-digit growth. UN medium projection: global population peaks near **10.3 billion in the mid-2080s** (UN World Population Prospects 2024).

6. Climate Change as the Defining Global Challenge

Anthropogenic climate change is the single most integrative 21st-century challenge. It is physical (forced by greenhouse-gas emissions from Topic 2.2 fossil combustion), biological (biodiversity loss), economic (trillions in stranded assets and adaptation cost), political (demands cooperation between >190 states), and ethical (those least responsible are most exposed). Long-term global warming has reached ~1.3 °C above the 1850–1900 baseline (WMO); 2024 became the first calendar year with mean temperature exceeding 1.5 °C (Copernicus), though the Paris threshold refers to a multi-decadal average rather than a single year.

Paris Agreement (2015): Legally binding international treaty adopted at COP21 in December 2015, in force since November 2016. Commits parties (almost every state; the USA's second withdrawal took effect in January 2026) to hold the increase in global mean temperature to "well below 2 °C" above pre-industrial levels and to pursue efforts to limit warming to 1.5 °C, through *Nationally Determined Contributions* (NDCs) updated every five years, supported by climate finance (a \$100 bn/year developed-country goal) and a periodic *global stocktake*.

Loss and damage: The residual harms from climate change that exceed the adaptive capacity of countries and communities — for example a Bangladeshi village permanently submerged by sea-level rise, or a Pacific atoll rendered uninhabitable. The term appears in Article 8 of the Paris Agreement; a dedicated *Loss and Damage Fund* was agreed in principle at COP27 (Sharm el-Sheikh, 2022) and operationalised at COP28 (Dubai, 2023), though pledges (~\$700 million by early 2024) fall far short of estimated need (hundreds of billions of dollars annually).

COP summits — recent milestones

- **COP21 Paris (2015):** adopted the Paris Agreement.
- **COP26 Glasgow (2021):** Glasgow Climate Pact — first explicit reference to "phase down" (not out) of unabated coal; methane pledge (>100 countries, 30% reduction by 2030).
- **COP27 Sharm el-Sheikh (2022):** agreement in principle on the Loss and Damage Fund; limited progress on mitigation.
- **COP28 Dubai (2023):** operationalised Loss and Damage Fund; first Global Stocktake found the world far off-track; "UAE Consensus" called for "transitioning away from fossil fuels" — the first COP text to name fossil fuels directly, although language (no "phase out") was contested.

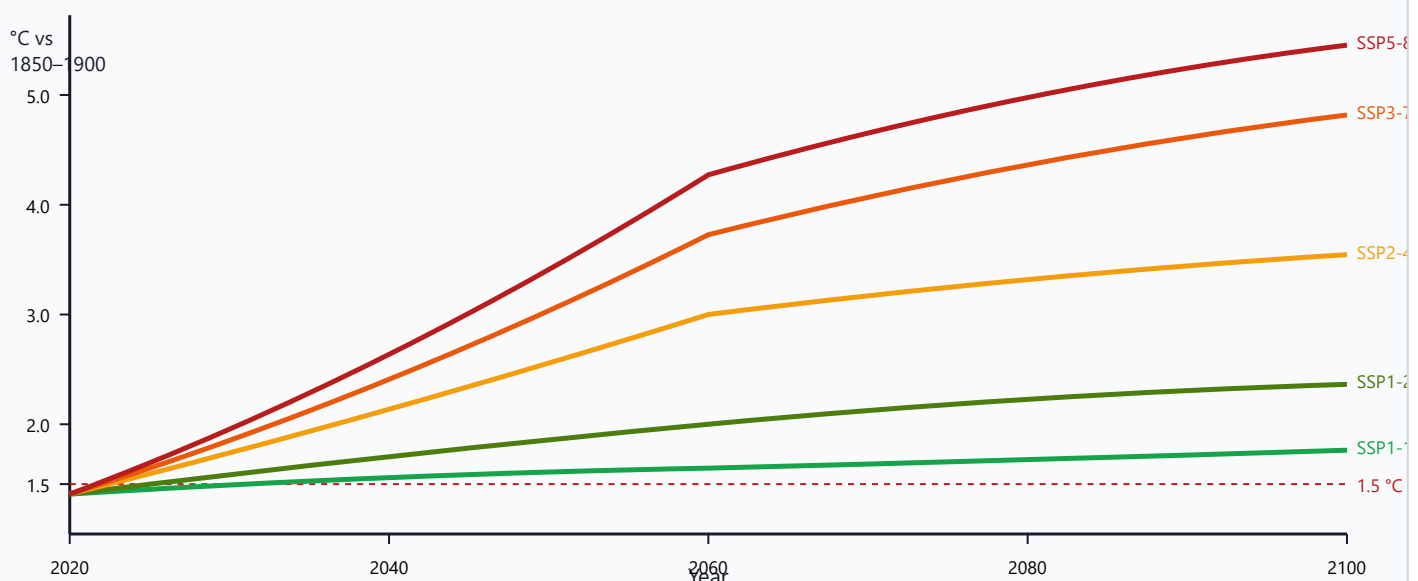
- **COP29 Baku (2024):** the “climate finance COP”; agreed a new collective quantified goal of \$300 bn/year by 2035 from developed countries — criticised as insufficient by the G77 and Climate Vulnerable Forum.
- **COP30 Belém (2025):** held in the Amazon; launched the Tropical Forest Forever Facility and agreed to triple adaptation finance by 2035, but the final decision contained no roadmap for moving away from fossil fuels.

Exam tip — keep COPs current: Eduqas examiners are impressed by candidates who know the *latest* COP. Memorise one-sentence summaries of COP26 to COP30 and use them to punctuate evaluations of global governance effectiveness.

Unequal impacts

Climate impacts are profoundly *uneven*. Small Island Developing States (SIDS) — Tuvalu, Kiribati, Marshall Islands, Maldives — face existential risk from sea-level rise (sea level has risen ~21 cm since 1900; rate now ~4.5 mm/yr). The Sahel (Mali, Burkina Faso, Niger, Chad) faces compounding heat, drought and conflict. Least Developed Countries (LDCs) contribute ~1% of cumulative emissions but bear disproportionate impacts. By contrast, Russia gains longer growing seasons and navigable Arctic routes in the short term. This *climate injustice* is the moral spine of the loss and damage debate.

IPCC SSP pathways — projected global warming to 2100



Climate feedbacks and non-linearity

A central reason climate change is hazardous is its non-linearity. Positive feedbacks amplify warming beyond the direct forcing from CO₂: as Arctic sea ice retreats, a dark ocean replaces reflective white ice (the *ice-albedo feedback*), absorbing ~90% of incoming shortwave radiation rather than reflecting ~80%. Thawing permafrost (~15 million km², storing ~1,500 GtC — roughly twice the carbon currently in the atmosphere) releases CO₂ and methane; the methane budget update by the Global Carbon Project (2023) showed atmospheric methane rising at the fastest rate on record. Boreal forest dieback, wildfire release of stored carbon (Canada's 2023 fire season released an estimated ~480 Mt of carbon (~1.8 Gt CO₂) — several times Canada's annual fossil-fuel emissions), and potential Amazon tipping toward savanna represent further feedbacks. The Earth system is not a linear machine whose response is proportional to input; it is a set of coupled non-linear systems with tipping points.

7. Food Security

The FAO's *State of Food Security and Nutrition in the World 2025* estimated that **673 million people** — 8.2% of the world's population — were undernourished in 2024, a second consecutive annual fall from the 2022 peak, but still well above the pre-pandemic level of 2019. The decline is driven by South and South-East Asia and South America; hunger is still *rising* across most of Africa (307 million undernourished, 20.2% of the continent) and in Western Asia. Drivers are what FAO calls the *three Cs*: *conflict* (Ukraine, Yemen, Sudan), *climate* (Horn of Africa drought), and *cost-of-living* (2022 energy and fertiliser price spike). Food insecurity is a synoptic issue par excellence — it depends on Topic 2.1 (water for irrigation), Topic 2.2 (energy for fertilisers and transport), Topic 2.3 (migration triggered by hunger), and global governance capacity.

Debates over **sustainable intensification** (produce more on existing land through precision agriculture, drought-tolerant varieties, drip irrigation) versus **land sparing** (intensify on small areas so that wider land is left to nature) versus **agroecology** (mixed systems with biodiversity at their core) are central to the 21st-century food policy agenda.

Food security is also a *trade* question. Just six countries (USA, Russia, Ukraine, Canada, Australia, Argentina) supply ~60% of global wheat exports; ten countries supply ~80% of maize. Concentration creates systemic vulnerability: when one producer fails (climate) or is cut off (war, export bans), the shock is global. The 2007–08 food-price crisis was exacerbated by export bans from India, Vietnam and Argentina; the 2022 crisis by Indian wheat and rice export restrictions. Food-system resilience therefore requires diversified

production, strategic reserves (China holds ~50% of global wheat reserves), and rules-based trade — all areas where the WTO's Agreement on Agriculture is under strain.

The **EAT-Lancet planetary health diet** (2019) proposed a reference diet that could feed 10 billion within planetary boundaries: roughly halving global red-meat and sugar consumption while doubling fruit, vegetable, legume and nut intake. Its implementation has been politically fraught; shifting diets intersects with culture, income and inequality. Food security in the 21st century therefore overlaps with public-health governance, climate policy, biodiversity conservation and trade law simultaneously — the classic synoptic tangle.

8. Water Security (link to Topic 2.1)

Water security is covered substantively in Topic 2.1 (water cycle and water insecurity) and is a synoptic case here. Key 2.5-relevant framings: ~2 billion people lack safely managed drinking water; ~2.3 billion live in water-stressed countries (UN-Water, 2023); 153 countries share 286 transboundary rivers and aquifers, requiring cooperation rather than conflict. "Water wars" are rarer than the media suggest; shared basins have produced >3,600 treaties since 805 CE (Oregon State TFDD). The Grand Ethiopian Renaissance Dam (GERD) dispute with Egypt and Sudan on the Nile, however, shows how tensions can escalate.

9. Energy Security and Transition (link to Topic 2.2)

Topic 2.2 covers the carbon cycle and the transition from fossil fuels. From a 2.5 perspective three points dominate:

- **Energy geopolitics:** Russia's 2022 invasion of Ukraine (see case study) weaponised gas exports, triggering an energy crisis in Europe; the EU reduced reliance on Russian gas from ~40% to ~15% of imports by the end of 2023.
- **Critical minerals:** the clean-energy transition is mineral-intensive. An electric vehicle contains ~6× more minerals than an internal-combustion car. The Democratic Republic of Congo mines ~70% of global cobalt; China refines ~60% of lithium and ~80% of rare earths. "Green geopolitics" is producing new dependencies.
- **Transition pace:** renewables added ~510 GW of capacity in 2023 (~50% above 2022; IEA); solar PV alone added more capacity than all fossil fuels combined. Yet fossil fuels still supplied ~80% of primary energy demand.

10. Ocean Governance (link to Topic 2.4)

Topic 2.4 covers UNCLOS, the International Seabed Authority, Regional Fisheries Management Organisations and the 2023 BBNJ “High Seas Treaty”. The synoptic relevance for 2.5 is that oceans exemplify the challenge of governing **global commons**: resources that nobody owns and everybody can over-exploit. Ocean governance rehearses the same problems — commitment, compliance, enforcement, finance — that beset climate, biodiversity and atmosphere governance.

Transition is also politically uneven. At one extreme, Denmark sourced >55% of electricity from wind in 2023; Norway’s new car sales were ~90% EV in 2024. At the other, coal consumption reached an all-time high globally in 2023 (~8.7 billion tonnes), driven by Asia. The International Energy Agency estimates that limiting warming to 1.5 °C requires tripling renewable capacity and doubling energy-efficiency improvements by 2030 (a commitment made at COP28) — a target widely viewed as physically feasible but politically very demanding. The geography of the energy transition therefore maps onto the geography of capital, technology, state capacity and political will — a synoptic tangle linking 2.2 directly to 2.5.

11. Health and Pandemics

COVID-19 (2020–2023) was the defining health event of the century so far. The World Health Organization reported ~7 million confirmed deaths; excess-mortality estimates are 2–3× higher. The pandemic demonstrated the *velocity of interdependence*: the SARS-CoV-2 virus reached every inhabited continent within ~10 weeks of the first reported cluster. Vaccine development at unprecedented speed (<1 year for mRNA vaccines) was an extraordinary scientific achievement; vaccine distribution was a governance failure — by mid-2021 ~80% of doses had gone to high- and upper-middle-income countries, despite the COVAX facility. New pandemic risks — bird flu H5N1 spillover, antimicrobial resistance, Ebola, Marburg — compound with biodiversity loss (zoonotic spillover correlated with habitat disturbance).

Beyond pandemics, global health challenges include the “double burden” of communicable and non-communicable disease facing middle-income countries (Mexico, Brazil, South Africa — combining HIV/TB and diabetes/obesity); maternal mortality (still ~260,000 deaths in 2023, WHO, concentrated in sub-Saharan Africa); and antimicrobial resistance (AMR), projected by the O’Neill review (2016) to kill ~10 million people annually by 2050 if unchecked. Universal Health Coverage (SDG3.8) is one of the most off-track SDGs. Climate change is also a health issue: heat deaths, expanded range of vector-borne disease (dengue now present in southern Europe), and disruption to health systems from extreme events are all intensifying. The 2023 Lancet Countdown tracked deterioration in 10 of 15 indicators of climate-health.

12. Technology and the Digital Divide

The ITU's 2023 estimate is that **~67% of the world's population uses the internet** — meaning ~2.6 billion people (~33%) remain offline. The divide is geographically concentrated: ~63% of Africa is offline versus ~11% of Europe. Within countries there is also a digital divide by income, age, gender and rurality. Beyond access, the 21st-century technology agenda includes:

- **AI disruption:** large-language-model deployment from 2022 (ChatGPT) to 2025 (frontier models in medicine, code, science); labour-market effects; governance through the EU AI Act (2024), the US Executive Order on AI (2023, revoked in January 2025) and the Bletchley Declaration (2023).
- **Cybersecurity:** ransomware, critical-infrastructure attacks (Colonial Pipeline 2021), state-sponsored hacking.
- **Data governance and surveillance:** GDPR (EU, 2018), China's social-credit experiments, Five Eyes intelligence sharing.
- **Platform power:** five US "Big Tech" firms + Chinese counterparts (Tencent, Alibaba, ByteDance) structure the informational environment of billions of people.
- **Digital geography:** submarine fibre-optic cables (~500 active, ~1.4 million km) carry ~99% of intercontinental data; their chokepoints (Red Sea, Strait of Malacca, Luzon Strait) are new geopolitical pressure points (three Red Sea cables were cut in February 2024, apparently by the dragging anchor of a Houthi-damaged ship).

The digital divide is itself a synoptic issue. Lack of connectivity in rural Africa and South Asia constrains participation in the global economy, access to education (stark during COVID school closures), remittance flows, climate-adaptation information and civic participation. Conversely, surveillance capitalism (Zuboff) and platform concentration raise questions about *too much* digital exposure in affluent societies. The governance response is fragmented: the EU is the clear global regulator via the "Brussels effect" (GDPR 2018, DMA/DSA 2022, AI Act 2024); the US relies on markets plus sectoral regulation; China maintains a separate sovereign internet with the Great Firewall and the "Cyberspace Administration". This *splinternet* pattern is itself a 21st-century governance feature.

13. Future Scenarios — the SSP Framework

The IPCC's **Shared Socioeconomic Pathways** (SSPs) are not forecasts but structured narratives used for climate, land, water and economic modelling. They pair socioeconomic storylines (SSP1–5) with radiative-

forcing pathways (e.g. "1.9", "2.6" W/m² by 2100) to generate scenarios used throughout AR6.

Pathway	Narrative	World in 2100	Warming range (°C)
SSP1 Sustainability	Shift to inclusive, sustainable development; strong cooperation; low material footprints	~7 bn people; high HDI; low emissions	1.4 (SSP1-1.9), 1.8 (SSP1-2.6)
SSP2 Middle of the road	Historical trends continue; moderate progress	~9 bn; uneven progress	2.7 (SSP2-4.5)
SSP3 Regional rivalry	Resurgent nationalism; fragmented blocs; low investment in health/education	~12.6 bn; high inequality	3.6 (SSP3-7.0)
SSP4 Inequality	Stratified societies: a small elite vs a fragmented majority	~9.3 bn; deep inequality	~2.5–3.5
SSP5 Fossil-fuelled development	Rapid global economic growth powered by fossil fuels; techno-optimism	~7.4 bn (late-century decline); high GDP	4.4 (SSP5-8.5)

The SSPs help examiners test candidates' ability to reason about *futures*. They also underpin the concept of **tipping points**: non-linear, potentially irreversible transitions in the Earth system (Amazon dieback, Greenland ice-sheet collapse, AMOC shutdown, permafrost thaw, coral reef die-off) whose probability rises with each fraction of a degree of warming.

Tipping points are not only ecological. *Social tipping points* — the rapid normalisation of a previously fringe behaviour once it passes a threshold share of a population — may also shape 21st-century trajectories, for good (the collapse of apartheid, the rapid rise of LGBTQ+ recognition in many societies, the sudden normalisation of EVs in Norway and China) or ill (the collapse of democratic norms under populist regimes). *Political tipping points* include state failure and the cascading effect on neighbours (Somalia, Syria, Venezuela). Raworth's *Doughnut Economics* (2017) reframes 21st-century development as a question of staying within planetary boundaries (the "ecological ceiling") while meeting social thresholds (the "social foundation") — an explicit departure from GDP-growth-centric models.

14. Global Governance Architecture

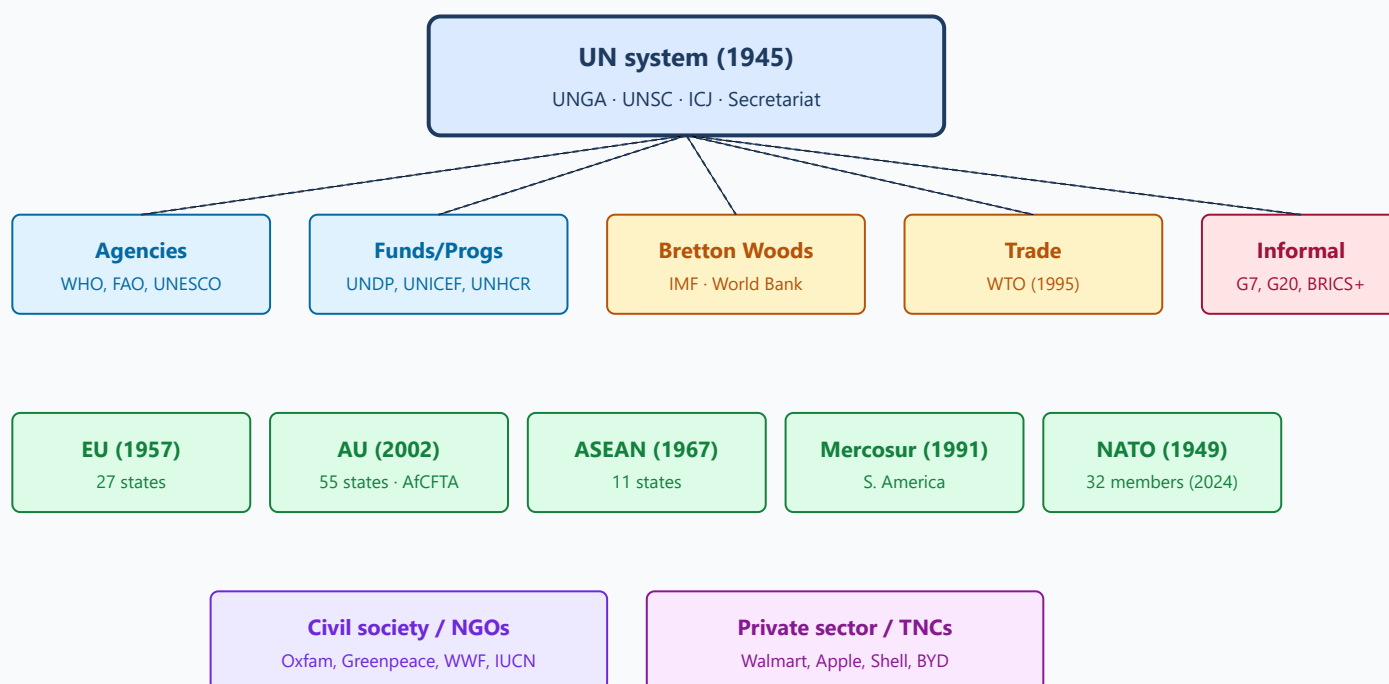
The “governance architecture” of the 21st century consists of overlapping, sometimes competing institutions at multiple scales.

UN system: The *United Nations Organization* (founded 1945, HQ New York) has 193 member states and comprises six principal organs: General Assembly, Security Council (5 permanent members with veto — P5 — plus 10 elected), ECOSOC, Secretariat, International Court of Justice, and the now-dormant Trusteeship Council. Specialised agencies (WHO, FAO, UNESCO, ILO), funds and programmes (UNDP, UNICEF, UNHCR, WFP) and related bodies (WTO, IAEA) constitute the wider “UN system”.

Body	Founded	Role	Key criticisms
UN General Assembly	1945	Deliberative forum, one-state-one-vote	Resolutions non-binding
UN Security Council	1945	International peace & security; binding decisions	P5 veto; 1945 geography; reform stalled
IMF	1944 (Bretton Woods)	Macroeconomic stability; lender of last resort	“Washington Consensus” conditionality
World Bank	1944	Development finance	US presidency convention; displacement effects
WTO	1995 (GATT 1947)	Trade rules; dispute settlement	Appellate body paralysed since 2019
G7	1975	Informal forum of advanced democracies	Excludes China, India
G20	1999 (leaders 2008)	Forum of systemically important economies	Consensus-based → lowest common denominator

European Union	1957 Treaty of Rome	Deepest regional integration; single market	Democratic deficit debates; enlargement fatigue
African Union	2002 (OAU 1963)	Continental body; AfCFTA trade area	Capacity, funding constraints
ASEAN	1967	SE Asian cooperation; “ASEAN way” consensus	Limited on South China Sea
Mercosur	1991	South American customs union	Political friction Argentina–Brazil
BRICS+	2009 (expanded 2024)	Non-Western coordination	Heterogeneous members; no binding rules

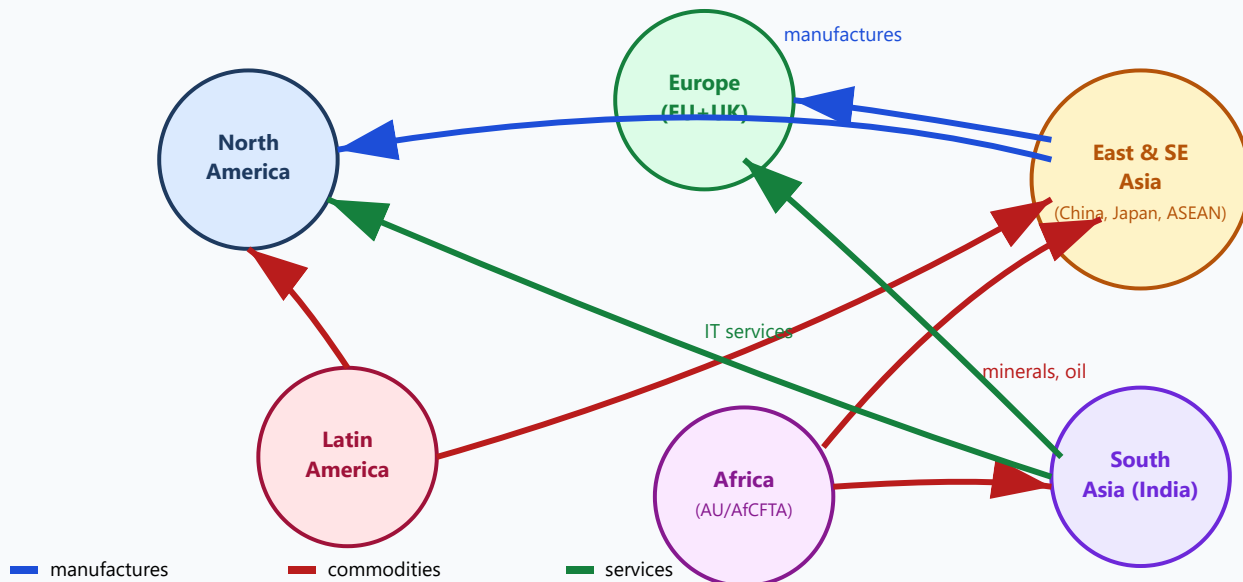
Global governance architecture (schematic)



Exam tip — evaluate, don’t describe: Eduqas 2.5 questions on global governance rarely reward description. Ask yourself three evaluative questions: (1) Is it *legitimate* (fair representation)? (2) Is it *effective* (produces outcomes)? (3) Is it *adaptive* (can reform)? Use these as paragraph topics in an essay.

15. Global Trade Flows — A Schematic View

Schematic 21st-century trade flows



16. Required Case Studies

Case Study 1 — COVID-19 pandemic (2020–2023): First cases reported in Wuhan, Hubei, China in December 2019; WHO declared a Public Health Emergency of International Concern (PHEIC) on 30 January 2020 and a pandemic on 11 March 2020. By mid-2024, the WHO recorded ~7.0 million confirmed deaths (true toll, including excess mortality, estimated 15–25 million). Global GDP contracted ~3.1% in 2020 — the sharpest peacetime contraction since the 1930s. Vaccine development was historically fast: Pfizer-BioNTech and Moderna mRNA vaccines received emergency use authorisation in December 2020, <12 months after genome sequencing. Yet distribution was starkly unequal — by mid-2021 over 80% of doses had been administered in high- and upper-middle-income countries, while the COVAX facility (WHO/CEPI/GAVI) delivered only ~1.1 billion doses against a target of 2 billion by end-2021; Africa's vaccination rate was still under 20% at that point. Long-term effects include permanent shifts to remote and hybrid work in knowledge-economy sectors, an estimated ~10% one-off loss to learning in affected cohorts (World Bank), supply-chain restructuring ("friendshoring", inventory build-up), and reinvigorated debates over WHO reform and a *Pandemic Agreement* (negotiated from 2021 and adopted by the World Health Assembly in May

2025, without the USA, whose withdrawal from the WHO took effect in January 2026). Synoptic connections: interdependence (2.5), global governance weakness (2.5), migration restrictions (2.3), carbon-emissions drop during lockdowns (2.2).

Case Study 2 — Russia-Ukraine war (February 2022–present): Russia's full-scale invasion of Ukraine on 24 February 2022 is the most consequential geopolitical event in Europe since 1945. Direct impacts: >500,000 estimated military casualties on both sides by early 2024 (Western intelligence estimates; Ukrainian civilian deaths >10,000 confirmed, UN OHCHR). Displacement: ~6.5 million Ukrainian refugees in Europe (UNHCR, 2024) — the largest refugee flow in Europe since WWII, primarily to Poland, Germany, Czechia (Topic 2.3 synoptic link). Energy crisis: Russia cut gas flows via Nord Stream; EU wholesale gas prices peaked at ~€340/MWh in August 2022 (~10× pre-crisis); the EU reduced Russian gas import share from ~40% to ~15% by end-2023, replaced by US LNG, Norwegian pipeline gas, Qatari LNG. Food crisis: Russia and Ukraine combined produced ~30% of global wheat exports and ~70% of sunflower oil pre-war; wheat prices rose ~50% in the first months of the war before the Black Sea Grain Initiative (July 2022–July 2023) restored partial exports. Geopolitical rupture: NATO enlargement (Finland joined April 2023, Sweden March 2024); unprecedented Western sanctions (~\$300 bn Russian central-bank reserves frozen); weakened UN Security Council (Russia's veto); deepened Russia–China partnership ("no limits" joint statement February 2022); Ukraine's EU candidate status granted June 2022. Synoptic: energy (2.2), migration (2.3), food security, global governance.

Case Study 3 — China's Belt and Road Initiative (BRI, 2013–present): Announced by Xi Jinping in Kazakhstan (September 2013) as the "Silk Road Economic Belt" and in Indonesia (October 2013) as the "21st-Century Maritime Silk Road". BRI now engages >140 countries accounting for ~65% of global population; cumulative Chinese engagement (investments + construction contracts) is estimated at ~\$1 trillion by 2023 (Green Finance & Development Center, Shanghai). Flagship projects include the China–Pakistan Economic Corridor (CPEC, ~\$62 bn; Gwadar Port); the Mombasa–Nairobi Standard Gauge Railway (Kenya, \$3.2 bn); Hungary–Serbia Belgrade–Budapest railway; Hambantota Port (Sri Lanka); the Jakarta–Bandung High-Speed Railway (Indonesia, opened October 2023). Key criticisms: "debt-trap diplomacy" (the 2017 Hambantota lease to a Chinese SOE for 99 years after Sri Lanka could not service loans is the canonical case, though researchers at Chatham House and the Rhodium Group show most BRI debt is renegotiated rather than seized); environmental impacts (coal-plant financing); governance and corruption. Post-2021 pivot: a "Green BRI" was announced at the September 2021 UN General Assembly (Xi committed China to stop financing overseas coal

power). Strategic response: the US and G7 launched *Build Back Better World* (B3W, 2021), relaunched as the *Partnership for Global Infrastructure and Investment* (PGII, 2022, targeting \$600 bn by 2027); the EU announced *Global Gateway* (€300 bn by 2027). Synoptic: superpower competition (2.5), energy transition (2.2), global governance (2.5).

Case Study 4 — Global food crisis 2022–2024: A compounded food crisis produced by the three Cs — conflict, climate, cost of living. Triggers: (a) Russia's invasion of Ukraine (February 2022) disrupted ~30% of global wheat exports and ~70% of sunflower-oil exports; (b) a multi-year drought in the Horn of Africa (five consecutive failed rainy seasons, 2020–23) (Ethiopia, Somalia, Kenya) driven by La Niña and climate change — the longest recorded in the region; (c) lingering pandemic effects on logistics; (d) the 2021–22 natural-gas price spike pushed fertiliser prices up ~3× (urea ~\$1,000/t peak 2022) because ~70% of fertiliser production cost is gas. Impacts: FAO Food Price Index peaked at 159.7 in March 2022, its highest on record; ~735 million undernourished in 2022 as first reported (FAO, SOFI 2023 — later revised down to ~695 million in SOFI 2025, which puts the 2024 figure at 673 million); acute food insecurity (IPC Phase 3+) affected ~282 million in 59 countries. Responses: the *Black Sea Grain Initiative* (brokered by Turkey and the UN, 22 July 2022) allowed ~33 million tonnes of Ukrainian grain to be exported before Russia withdrew in July 2023; the World Food Programme (WFP) — which itself sources ~50% of its grain from Ukraine — faced unprecedented funding shortfalls (raised ~\$14 bn in 2022 against ~\$23 bn assessed need). Synoptic: conflict (2.5), climate (2.2), energy (2.2), migration and displacement (2.3).

Case Study 5 — COP27–28 “loss and damage” pathway: After more than two decades of campaigning by the Alliance of Small Island States (AOSIS) and the G77 grouping, **COP27 (Sharm el-Sheikh, November 2022)** produced the breakthrough “Sharm el-Sheikh Implementation Plan” with an agreement *in principle* to establish a dedicated Loss and Damage Fund for vulnerable developing countries. A Transitional Committee was tasked with operationalisation. **COP28 (Dubai, November–December 2023)**, presided over by Sultan Al Jaber (simultaneously CEO of Abu Dhabi National Oil Company), operationalised the Fund on Day 1 — a procedural milestone — with initial pledges totalling ~\$700 million (UAE \$100m, Germany \$100m, UK £60m, US \$17.5m, Japan \$10m). Critics including the Climate Action Network observed that the figure was <0.2% of the hundreds of billions in annual climate-driven damages in developing countries. COP28 also produced the first *Global Stocktake* under the Paris Agreement, confirming the world is far off-track (current NDCs imply warming of ~2.5–2.9 °C by 2100) and adopting the “UAE Consensus” text calling for “transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner” — the first explicit

naming of fossil fuels in a COP decision, although the weaker language (“transitioning away” rather than “phase out”) was contested by AOSIS. Synoptic: climate (2.2), global governance effectiveness (2.5), North–South equity, ocean-linked impacts (2.4).

Shorter supporting case studies

Beyond the five required major cases, Eduqas candidates benefit from a portfolio of smaller *place-specific* examples that can be dropped into 2.5 essays.

- **Tuvalu — digital statehood:** in November 2022, the Tuvaluan Foreign Minister Simon Kofe proposed the world’s first “digital nation”, preserving cultural heritage and statehood in the metaverse in anticipation of physical submergence. By 2024 Tuvalu’s constitutional amendments formally asserted the persistence of statehood even without territory.
- **Rwanda — developmental state model:** GDP growth averaged >7% in the 2010s; state-led, tech-forward (Kigali Innovation City), UK migration “offshoring” deal (2022–2024, ruled unlawful by the UK Supreme Court November 2023).
- **Qatar — soft-power spending:** 2022 FIFA World Cup (~\$220 bn infrastructure); Al Jazeera; Qatar Airways as a state geopolitical instrument; mediator in the Israel-Hamas ceasefire negotiations 2023–24.
- **Taiwan — semiconductor chokepoint:** TSMC produces ~90% of the world’s most advanced chips (<5nm). A single island supplies the informational backbone of the global economy — a geopolitical risk visible in US CHIPS Act (2022) and Biden-Xi tensions.
- **Pacific climate migration:** Australia’s 2023 “Falepili Union” with Tuvalu offered permanent-residency pathways for ~280 Tuvaluans annually — an early template for treaty-based climate mobility.
- **Wales (UK) — Well-being of Future Generations Act:** as noted, constitutionally embedded SDG-style goals; the Future Generations Commissioner has statutory authority to challenge public-body decisions.
- **Gaza, 2023–2024:** Israel-Hamas war since October 2023; ~2.3 million population; >40,000 reported deaths by mid-2024 and over 67,000 by the October 2025 ceasefire (Gaza Ministry of Health); ICJ “plausible genocide” ruling January 2024. Illustrates both acute humanitarian crisis and the limits of global governance (UN vetoes).
- **Sudan, 2023–present:** civil war between SAF and RSF from April 2023; >10 million displaced — the world’s largest displacement crisis; famine formally declared in Zamzam camp, August 2024; the RSF captured El Fasher in October 2025.

Case study synthesis table

Case	Physical / human	Scale	Governance response	Synoptic links
COVID-19	Biological (zoonotic) → human	Global	WHO, COVAX, national lockdowns	2.3, 2.5
Russia–Ukraine	Human (political)	Regional → global	UN condemnation, NATO, sanctions	2.2, 2.3, food
Belt & Road	Human (economic)	Trans-regional	Chinese bilateral & AIIB; PGII response	2.2, 2.5
Food crisis 2022–24	Coupled physical + human	Global	FAO, WFP, Black Sea Initiative	2.1, 2.2, 2.3
COP27–28 loss & damage	Climate (physical) → social	Global	UNFCCC Fund operationalised	2.2, 2.4

17. Cross-Cutting Theme — Demography

Demographic change is the slowest-moving but most predictable 21st-century transformation. Four patterns dominate.

- **Continued population growth in sub-Saharan Africa:** from ~1.15 bn (2023) to ~2.1 bn by 2050 (UN medium variant). Niger's total fertility rate was still around 6 in the early 2020s.
- **Ageing in Europe, East Asia, North America:** Japan's population peaked in 2008 and is projected to fall from ~124m to ~87m by 2070. China's population fell in 2022 for the first time since the 1961 famine.
- **India overtaking China as the most populous country (2023):** a symbolic shift with implications for UN Security Council reform debates.
- **Global population peak < 11 bn:** most demographers now expect peak before 2100. This is quite different from the Malthusian predictions that dominated 20th-century debate.

Demographic dividend: The economic growth potential that arises when a country's working-age population is unusually large relative to its dependents — the stage at which fertility has fallen but the pre-fall cohort is still in the labour market. India, Nigeria, Ethiopia and Bangladesh are entering this window; East Asia already banked it in the 1960s–1990s; Europe has exited it.

Demography interacts powerfully with every other 21st-century challenge: with climate (Africa's emissions profile as it industrialises), with migration (ageing-driven demand in the North vs growth-driven supply in the South), with inequality (young unemployed populations and political instability), with food security (feeding 10 billion), and with global governance (a UNSC still structured around 1945 demographic weights).

18. Cross-Cutting Theme — Urbanisation

More than **56% of humanity** (~4.5 billion people) now live in urban areas; UN DESA projects ~68% by 2050. Almost all population growth to 2050 will be urban and most of it will be in Asia and Africa. Cities are simultaneously engines of 21st-century development and concentrations of 21st-century risks.

- **Megacities:** >10 million population; the UN's 2025 World Urbanization Prospects, using a new harmonised city definition, counts 33 in 2025, led by Jakarta (~42m), Dhaka (~40m) and Tokyo (~33m); older UN estimates ranked Tokyo first.
- **Informal settlements / slums:** ~1.1 billion urban residents live in slums or inadequate housing (UN-Habitat). Kibera (Nairobi), Dharavi (Mumbai), Orangi (Karachi), Neza (Mexico City) are canonical cases.
- **Urban climate risk:** >800 million urban residents are exposed to coastal flood risk (Hallegatte et al., 2013); a 1-metre sea-level rise by 2100 would threaten Jakarta, Ho Chi Minh City, Bangkok, Lagos, Miami.
- **Smart cities and digital twins:** Singapore, Barcelona, Helsinki are experimenting with data-driven urban governance; ethical concerns about surveillance.
- **Right to the city (Lefebvre):** an influential normative framing for urban-justice movements from Bogotá to Barcelona.

Urbanisation is a synoptic nexus: it links climate (heat islands, emissions), water (2.1), energy (2.2), migration (2.3), inequality and governance. Many of the 21st-century's crucial policy choices will be urban choices.

19. Cross-Cutting Theme — Biodiversity Loss

Biosphere integrity is one of the planetary boundaries already transgressed (Rockström et al. update, 2023). WWF's Living Planet Index 2024 reported an average ~73% decline in monitored vertebrate populations since 1970. Drivers in descending order: land-use change (agriculture & forestry), direct exploitation (hunting, fishing), climate change, pollution, invasive species. **IPBES 2019** estimated ~1 million species threatened with extinction.

Governance responses:

- **Convention on Biological Diversity (CBD, 1992, Rio):** 196 parties; the US has not ratified.
- **Kunming-Montreal Global Biodiversity Framework (GBF, COP15, December 2022):** 23 targets for 2030, including the headline "30 by 30" (protect 30% of land and sea by 2030) and a pledge to mobilise \$200 bn/year for biodiversity by 2030.
- **CITES (1973):** governs trade in endangered species; 184 parties.
- **BBNJ Treaty (2023):** high-seas marine protected areas (Topic 2.4 link).
- **IUCN Red List:** scientific authority on extinction risk.

Biodiversity and climate are increasingly treated as twin crises: a degraded biosphere sequesters less carbon; climate change shifts ecological zones faster than many species can adapt. The *nature-based solutions* agenda — mangrove restoration, peatland protection, reforestation — attempts to solve both together.

Biodiversity crises are highly *located*. Hotspots include:

- **Amazon basin:** ~17% deforested; a 2021 *Nature* paper (Gatti et al.) warned that the eastern Amazon is now a net carbon source.
- **Coral Triangle:** Indonesia–Philippines–Solomon Islands; highest marine biodiversity on Earth; vulnerable to bleaching (fourth global mass bleaching event declared April 2024).
- **Congo Basin:** second-largest tropical forest; under pressure from mining (cobalt, coltan), logging, agriculture.
- **Madagascar:** ~80% of species endemic; severe habitat loss.
- **East African rift lakes:** cichlid fish diversity; pollution and invasive species.

20. Evaluating Global Governance

A robust evaluation of 21st-century global governance requires distinguishing three criteria.

- **Effectiveness** — does the regime change behaviour? The Montreal Protocol (1987) on ozone is the gold-standard success (99% compliance, ozone layer projected to recover by the 2060s). The Paris Agreement is weaker: no enforcement, but NDCs have shifted expectations (central estimate moved from ~3.5 °C to ~2.7 °C since 2015). The WTO's Appellate Body has been paralysed since December 2019 because the US blocks appointments.
- **Legitimacy** — is the process perceived as fair? The UN Security Council's 1945 geography (five P5 vetoes; no African or Latin American permanent member) is the most obvious legitimacy deficit. Bretton Woods conventions (US president of the World Bank, European head of the IMF) are criticised. BRICS and the G77+ bloc increasingly challenge this asymmetry.
- **Adaptive capacity** — can the regime evolve? The 2023 BBNJ (High Seas) Treaty is evidence of UNCLOS evolving (see Topic 2.4); the Pandemic Treaty and WHO IHR reforms are tests of whether the WHO can adapt after COVID. The UN Security Council reform process has been stalled since 2005 — evidence of adaptive failure.

Exam tip — use the “effectiveness · legitimacy · adaptive capacity” triad: Deploy this three-part evaluation when the question is “assess global governance of X”. Each of the three becomes a paragraph with evidence. You will almost always reach a nuanced conclusion: governance is partial — effective in some domains, legitimate in some respects, adaptive in some areas — and the distribution of these matters more than a single grade.

Exam tip — structuring the 30-mark Section C essay: (1) *Introduction* — define key terms, state a clear thesis, outline scope. (2) *Three evaluative paragraphs* — each with claim, case study, counter-evidence, mini-conclusion. (3) *Synoptic bridge paragraph* — draw explicit links across topics (e.g. climate → migration → food). (4) *Conclusion* — return to thesis, qualify it with scale/temporal caveats. Use signposting: “furthermore”, “however”, “conversely”, “overall”.

Exam tip — avoid the “everything-is-terrible” trap: Weaker essays paint a uniformly bleak picture. Stronger essays acknowledge progress — global extreme poverty fell from ~38% of world population in 1990 to ~9% by 2019 (World Bank); child mortality halved since 2000; the ozone layer is recovering; renewable capacity is scaling faster than predicted. Balanced evaluation earns higher band credit than either boosterism or doom.

Non-state actors in global governance

It is a mistake to treat global governance as exclusively interstate. A rich ecosystem of non-state actors shapes outcomes:

- **International NGOs:** Oxfam, Amnesty International, Human Rights Watch, Greenpeace, WWF, Médecins Sans Frontières. Provide research, advocacy, and service delivery; ~10 NGOs deliver the majority of humanitarian aid.
- **Transnational advocacy networks (Keck & Sikkink, 1998):** the climate movement (Fridays for Future from 2018, XR from 2018), debt-relief coalitions (Jubilee 2000), the anti-landmines coalition that produced the 1997 Ottawa Treaty.
- **Philanthropic foundations:** Gates Foundation (~\$70 bn endowment, GAVI, polio eradication), Bloomberg Philanthropies (C40 Cities), Open Philanthropy.
- **Epistemic communities:** IPCC, IPBES, the medical community around the WHO. Expert networks shaping consensus.
- **Cities and sub-national governments:** C40 Cities (97 mayors representing ~700m people), Under2 Coalition of sub-national leaders, the “America Is All In” (formerly “We Are Still In”) coalition of US states, cities and businesses during the 2017–21 federal climate withdrawal.
- **Private sector:** TNCs set de facto global standards (Walmart’s supply-chain sustainability rules, Apple’s privacy defaults); net-zero pledges by firms covering ~80% of global market capitalisation; corporate climate reporting regimes (TCFD, ISSB 2023).
- **Indigenous peoples:** ~6% of global population but stewards of ~80% of remaining biodiversity (UN); increasingly recognised in climate and biodiversity treaties as rights-holders, not just stakeholders.

The resulting “polycentric” (Ostrom) or “regime-complex” (Keohane and Victor) structure means that progress on any 21st-century challenge typically requires alignment across multiple levels, not just among states at the UN.

21. The Future of the UK in a Changing World

Eduqas is WJEC’s brand for England; UK (and Welsh) material is useful located context for Section C answers. Key points:

- **Post-Brexit trade:** the UK formally left the EU on 31 January 2020; the Trade and Cooperation Agreement (TCA, 1 January 2021) preserves tariff-free goods trade but imposes customs paperwork.

OBR estimates a ~4% long-run productivity hit versus counterfactual EU membership. New trade deals: UK–Australia (2021), UK–New Zealand (2022), CPTPP membership (December 2024), UK–India (signed July 2025).

- **“Global Britain”**: the 2021 and 2023 Integrated Reviews repositioned UK foreign policy with an “Indo-Pacific tilt”; AUKUS (September 2021) is the flagship security partnership with Australia and the USA.
- **Climate leadership**: the UK was first major economy to legislate net zero by 2050 (2019 amendment to Climate Change Act 2008); hosted COP26. But the Climate Change Committee (CCC) judged UK policy “significantly off-track” for its 2030 NDC in its 2024 progress report; the 2023 delay of the petrol/diesel car ban to 2035 was reversed in 2025, when the government restored a 2030 end-date for new pure petrol and diesel cars (hybrids allowed until 2035).
- **Wales specifically**: the *Well-being of Future Generations (Wales) Act 2015* obliges public bodies to consider long-term impacts, making Wales one of the first countries to put SDG-style goals into law. Wales declared a climate emergency in April 2019. Welsh Government targets net zero for its public sector by 2030.
- **Commonwealth relations**: 56 member states, ~2.5 bn people, but the institution’s political weight has arguably waned; post-Elizabeth II succession posed legitimacy questions.
- **Development and aid**: the UK merged DFID into the FCO in 2020 to form the FCDO, and cut the aid budget from 0.7% to 0.5% of GNI in 2021 (with a further cut to 0.3% by 2027 announced in February 2025) — moves criticised by the House of Commons International Development Committee and by NGOs such as Bond and Oxfam. In-country aid spending on refugees (counted as ODA under OECD rules) now consumes ~29% of the shrunken budget, reducing what reaches low-income countries.
- **Defence and security**: UK commitment to spend 2.5% of GDP on defence by 2027 (February 2025), plus the 2025 NATO pledge to reach 5% of GDP on defence and security by 2035; continued role in NATO, AUKUS, Joint Expeditionary Force; nuclear deterrent renewal (Dreadnought-class submarines).

The UK is therefore simultaneously a mid-sized power navigating a multipolar world, a climate leader in rhetoric but slipping in policy, and a test case for whether an old-industrial economy can re-invent itself through finance, services, low-carbon technology and education exports. For Eduqas candidates, the UK/Wales can be deployed as a *case-within-a-case* in any 2.5 essay.

22. Theoretical Frames for 21st-Century Geography

Eduqas examiners reward explicit use of *concepts* from human geography. The following frames are frequently useful in 2.5 essays.

- **Globalisation theory (Held, McGrew):** hyperglobalists see a borderless world, sceptics emphasise continued state power, transformationalists argue that globalisation transforms but does not abolish the state.
- **World-systems theory (Wallerstein):** core, semi-periphery, periphery; an economic hierarchy reproduced through the structure of trade.
- **Dependency theory (Frank, Cardoso):** “the development of underdevelopment” — peripheral economies are kept poor by their relationship with core economies, not by some internal deficit.
- **Modernisation theory (Rostow):** linear stages from traditional to high-mass-consumption; heavily criticised but still influential in aid practice.
- **Planetary boundaries (Rockström et al., 2009, updated 2023):** nine Earth-system processes (climate change, biosphere integrity, land-system change, freshwater change, biogeochemical flows, ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, novel entities). Six were transgressed in the 2023 update, and the 2025 Planetary Health Check added ocean acidification, making seven.
- **Doughnut economics (Raworth, 2017):** combines planetary boundaries with a social foundation (12 dimensions of well-being) to define a safe and just operating space for humanity.
- **Anthropocene:** proposed geological epoch recognising humanity’s global influence on the Earth system; the Anthropocene Working Group proposed the mid-20th-century “Great Acceleration” as a start date, though the International Commission on Stratigraphy declined formal ratification in 2024.
- **Risk society (Beck, 1986):** modernity has become defined by the production and management of manufactured risks (nuclear, financial, climate, digital) that exceed national boundaries.
- **Cosmopolitanism (Held):** the moral community is humanity, not the nation — a normative frame for global-justice arguments.

23. Contested Keywords and Concepts for 21st-Century Essays

High-band essays are fluent in *contested* terms — terms which different actors use to mean different things, and whose definition is itself politically loaded.

- **“Development”:** economic growth? capability expansion? freedom? well-being? Each definition implies different policies.
- **“Sovereignty”:** Westphalian (absolute territorial control)? Pooled (EU-style)? Responsible (R2P, 2005)? Digital? Climate-constrained?

- **“Security”**: traditional military? human security (UNDP 1994: economic, food, health, environmental, personal, community, political)? climate security? biosecurity?
- **“Just transition”**: who pays the cost of decarbonisation? ILO framework emphasises workers and communities dependent on fossil fuels.
- **“Resilience”**: bouncing back (engineering) or bouncing forward (transformative)? Criticised as depoliticising by some geographers (Hayes 2014).
- **“Net zero”**: gross emission cuts or offsetting? GHG Protocol Scope 1/2/3 distinctions matter; “science-based targets” (SBTi) attempt rigour.
- **“Climate justice”**: distributive (who bears costs), procedural (who gets a seat at COP), recognitional (which worldviews count, including indigenous perspectives).

24. Useful Data Snapshots for Quick Recall

Metric	Value	Source/year
World population	~8.1 billion	UN DESA, 2024
Projected peak population	~10.3 bn, mid-2080s	UN medium variant 2024
Global GDP (nominal)	~\$106 tn	IMF 2024
Global extreme poverty (\$2.15/day)	~9% of population	World Bank 2023
International migrants	~281 million	UN DESA 2020
Forcibly displaced persons	~120 million	UNHCR mid-2024
Remittances to LMICs	~\$656 bn	World Bank 2023
Global CO ₂ emissions	~37 Gt	GCP 2023
Temperature anomaly (2024)	~1.5 °C above pre-industrial	Copernicus 2024
Sea-level rise rate	~4.5 mm/yr	NASA 2024

Ocean acidification	~30% more acidic than 1750	IPCC AR6
Undernourished	~673 million (2024)	FAO SOFI 2025
Without internet	~2.6 billion	ITU 2023
Wealth held by dollar millionaires (richest ~1.5% of adults)	48.4%	UBS Global Wealth Report 2025
UN member states	193	UN
BRICS+ share of world GDP (PPP)	~35%	IMF 2024
Renewables share of electricity	~30%	IEA 2023
Coal consumption	~8.7 Gt (record)	IEA 2023

25. Essay Plans for Typical 2.5 Questions

Q: “To what extent is climate change the most serious 21st-century challenge? (30)”

- **Introduction:** define “serious” (scale, reversibility, spatial reach, speed); flag synoptic approach; state thesis (e.g. climate is the most structurally serious because it interlocks with all other challenges).
- **Para 1 — climate as global integrator:** IPCC AR6 evidence; 1.5 °C breach 2024; tipping points; SIDS; SSPs.
- **Para 2 — counter-challenges:** pandemics (COVID-19 case), conflict (Ukraine), inequality (dollar millionaires, ~1.5% of adults, hold 48% of global wealth — UBS 2025), AI disruption. Argue they may be faster or more immediate.
- **Para 3 — interlock:** climate amplifies all others (migration, food, conflict) — synoptic argument for climate as meta-challenge.
- **Conclusion:** yes in *structural* terms, but evaluated against *immediacy* other challenges may be as pressing. Qualified thesis.

Q: “Assess the effectiveness of global governance in managing 21st-century challenges. (30)”

- **Introduction:** define global governance (not just UN); introduce effectiveness·legitimacy·adaptive triad.
- **Para 1 — successes:** Montreal Protocol, Paris Agreement narrowing the warming range, BBNJ treaty, SDGs as coordinating frame.
- **Para 2 — failures:** Ukraine and UNSC paralysis; COVAX; WTO Appellate Body; climate finance shortfall.
- **Para 3 — mixed:** L&D fund operationalised but tiny; BRICS+ as challenger; NGO and private-sector roles.
- **Conclusion:** governance is not an on/off switch; it works unevenly across issue areas; argue for reform rather than dismissal.

Q: “To what extent are superpowers in decline? (30)”

- **Introduction:** define superpower (hard/soft/structural); note the question’s implicit focus on the US.
- **Para 1 — evidence of US relative decline:** share of world GDP (~40% 1960 → ~25% 2023); PPP overtaken by China; manufacturing dominance lost; WTO paralysis; Iraq and Afghanistan defeats.
- **Para 2 — evidence of persistent US dominance:** dollar ~88% of FX; largest military spending; AI frontier; tech-platform global reach; universities; USMCA; alliances (NATO, AUKUS).
- **Para 3 — rise of challengers:** China (BRI case study), India demographics, BRICS+, EU as regulatory power.
- **Conclusion:** shift from unipolar to multipolar — US relatively declining but not absolutely; rivals “rising” unevenly; structural power remains largely US/Western.

Q: “Evaluate the statement that the Brandt Line is no longer useful. (30)”

- **Introduction:** origins of Brandt Line; purpose as simplification.
- **Para 1 — reasons it persists:** still captures aggregate HDI gap; most LDCs south of the line; cumulative emissions asymmetry; historical relations (colonialism).
- **Para 2 — reasons it misleads:** China, India, Brazil rising; Gulf wealth; intra-national inequality (e.g. south Wales).
- **Para 3 — alternatives:** Rosling levels, World Bank income groups, world-systems, Doughnut.
- **Conclusion:** useful as heuristic with explicit caveats; dangerous if treated as literal description.

Q: “Discuss the view that migration is both a cause and a consequence of 21st-century challenges. (30)”

- **Introduction:** define migration types (Topic 2.3); acknowledge the two-way relationship; flag synoptic scope.
- **Para 1 — migration as consequence:** Ukraine refugees (war), Syria (conflict + climate drought), Venezuela (economic collapse), Sahel (climate + conflict), Sudan.
- **Para 2 — migration as cause:** remittances driving development; demographic balance in ageing states; brain drain weakening source countries; xenophobic backlash reshaping politics (Brexit, AfD).
- **Para 3 — feedback loops:** climate displacement + political reaction + weaker multilateralism; positive loops (diaspora investment).
- **Conclusion:** migration is constitutive of 21st-century geography, not peripheral.

Q: “Assess the statement that global challenges demand global solutions. (30)”

- **Introduction:** define “global” challenge vs locally concentrated challenge; define solution scales.
- **Para 1 — yes, global:** climate (atmosphere), pandemics (borders porous), oceans (commons), AI (cross-border).
- **Para 2 — but local implementation:** Paris NDCs nationally set; cities drive emissions cuts; agroecology locally adapted.
- **Para 3 — “glocal” governance:** Ostrom’s polycentric systems; subsidiarity; C40; indigenous stewardship.
- **Conclusion:** not a dichotomy — global frameworks + local implementation. Pure globalism produces legitimacy deficits; pure localism cannot tackle truly global commons.

26. Key Dates Timeline

Year	Event	Relevance
1944	Bretton Woods conference	IMF + World Bank created
1945	UN founded (San Francisco)	Post-war order
1948	Universal Declaration of Human Rights	Normative basis for governance

1980	Brandt Report	North-South framing
1987	Montreal Protocol	Gold-standard environmental treaty
1992	Rio Earth Summit (UNCED)	UNFCCC, CBD agreed
1995	WTO founded	Rules-based global trade
2000	Millennium Development Goals	Precursor to SDGs
2001	China joins WTO	Hyperglobalisation accelerates
2008	Global Financial Crisis	G20 to leaders level
2013	Belt and Road launched	Chinese geoeconomics
2015	SDGs adopted; Paris Agreement	Twin framework
2019	COVID-19 emergence	Pandemic era
2022	Russia invades Ukraine	Geopolitical rupture
2022	COP27: Loss & Damage agreed	Climate justice
2022	Kunming-Montreal GBF (COP15)	30 by 30 biodiversity target
2023	BBNJ High Seas Treaty	Ocean governance (2.4)
2023	COP28 Dubai: fossil fuel transition language	UAE Consensus
2024	First calendar year > 1.5 °C	Single-year breach; long-term average still below 1.5 °C
2024	COP29 Baku: \$300 bn finance goal	Inadequate but headline
2025	COP30 Belém; WHO Pandemic Agreement adopted	Mixed progress on global governance

27. Synoptic Revision Checklist

1. I can define globalisation and distinguish its economic, financial, political, cultural, environmental and technological dimensions.
2. I can critique the Brandt Line and give three specific examples of how it misrepresents the 21st-century world.
3. I can list at least five dimensions in which superpowers are compared, and I know the current rough values for US, China, EU, India and Russia.
4. I can name and apply HDI, Gini coefficient, MPI and GII, and state their limitations.
5. I know the 17 SDGs by theme and can map each to a topic on this specification.
6. I can describe the structure of the Paris Agreement (NDCs, 5-year stocktake, \$100 bn finance goal) and the outcomes of COP26 to COP30.
7. I understand “loss and damage” and can evaluate the Fund operationalised at COP28.
8. I can distinguish the five SSP pathways by narrative and by end-century warming.
9. I can draw the UN system and locate Bretton Woods, regional blocs and informal forums within it.
10. I have full notes on the five required case studies: COVID-19, Russia–Ukraine, BRI, food crisis 2022–24, COP27–28 loss & damage.
11. I can deploy the effectiveness · legitimacy · adaptive-capacity triad in an essay.
12. I can locate UK and Wales in the global picture (Brexit, Global Britain, Well-being of Future Generations Act).

Final exam tip — the synoptic reflex: Whatever 2.5 question you are facing, run a mental checklist: which topics (2.1–2.4) does this touch? Which scales (local, regional, national, global)? Which case studies can I evidence with? Which stakeholders (states, TNCs, NGOs, citizens, the non-human environment)? Which evaluation criterion (effectiveness, legitimacy, adaptive capacity)? Even one or two of these threads, explicitly pulled, moves your answer from Band 2 into Band 3/4 territory.

28. How the Topics Interlock — A Synoptic Map

The following is a shorthand map of how Topic 2.5 sits on top of 2.1–2.4. Use it as a mental prompt in the exam.

- **2.1 Water cycle & insecurity → 2.5:** transboundary water, GERD, water scarcity driving migration, climate intensification of droughts and floods, water for food security, water for energy cooling.
- **2.2 Carbon cycle & energy security → 2.5:** the central driver of climate change; Russia/Ukraine energy weaponisation; critical minerals; net-zero pledges; just transition.
- **2.3 Global migration → 2.5:** refugees as governance indicator; remittances as development driver; ageing-demand feedback; political backlash shaping multilateralism.
- **2.4 Global governance of oceans → 2.5:** template for global-commons governance; UNCLOS + ISA + RFMOs model for climate/biodiversity; BBNJ 2023 as recent breakthrough; loss and damage for SIDS.

Conversely, 2.5 gives you the *framework* (globalisation, superpowers, SDGs, governance architecture, futures) that contextualises the other four topics. The best Eduqas answers move fluently in both directions.

29. Glossary of Frequently-Tested Terms

Term	One-line definition
Globalisation	Deepening cross-border integration of economies, cultures, politics, environments.
Interdependence	Mutual dependence such that events in one place materially affect others.
Superpower	State able to project power globally on multiple dimensions.
Hegemon	Dominant power setting systemic rules (Gramsci; Cox).
Multipolar	World with 3+ roughly comparable power centres.
Geopolitics	Study of how geography shapes political power.
Global commons	Resource domain beyond national jurisdiction.
Paris Agreement	2015 climate treaty; 1.5 °C aspiration; NDCs; stocktakes.
NDC	Nationally Determined Contribution — each party's climate plan under Paris.

Loss and damage	Harms exceeding adaptive capacity; dedicated fund operationalised COP28.
SDGs	17 Sustainable Development Goals, 2015–2030.
HDI	Composite index: life expectancy + education + income (UNDP).
Gini coefficient	Measure of inequality, 0–1.
BRICS+	Brazil, Russia, India, China, South Africa + 2024 additions.
BRI	Belt and Road Initiative, China's infrastructure diplomacy since 2013.
Planetary boundaries	Nine Earth-system processes framing a safe operating space.
Anthropocene	Proposed epoch of human geological influence.
Tipping point	Non-linear, potentially irreversible Earth-system transition.
Polycentric governance	Overlapping authorities at multiple scales (Ostrom).
SSP	Shared Socioeconomic Pathway (IPCC scenario framework).